

Assignment no. 8

use the templates from Moodle for the Exercises!!

Exercise 1 LangChain Multi-Step Chains: Research Paper Analysis Pipeline

Complex tasks can often be decomposed into multiple smaller reasoning steps. In this exercise, you will use LangChain to construct a multi-step pipeline that analyzes a scientific research paper and produces management-ready output.

The goal is to transform a technical PDF into an executive summary and actionable recommendations.

Tasks:

1. Find a paper that about at topic that is relevant (Maybe some of [these](#)?).
2. Load and process the research paper in PDF format.
3. Build a multi-step LangChain pipeline that:
 - extracts the main research question and methodology,
 - identifies key findings,
 - generates a non-technical executive summary,
 - derives concrete action items and potential risks.
4. Ensure that each step of the pipeline produces an explicit intermediate output.
5. Reflect on the benefits of chaining multiple steps compared to using a single, large prompt.

Deliverables: A notebook containing the multi-step chain, intermediate outputs, and final executive summary with action items.

Exercise 2 LangChain Agents & MCP Tooling: AI Project Manager Agent

LangChain agents can autonomously decide which tools to use in order to achieve a goal. In this exercise, you will build an AI project manager agent that interacts with an existing MCP server for creating, editing, and querying tickets.

The agent should support basic project coordination tasks such as breaking down goals into tickets and updating their status.

Tasks:

1. Finalize MCP ticket server (if not already done in the exercise) to provide creating, editing, and querying tickets. ✓
2. Integrate the MCP ticket server as a tool usable by a LangChain agent.
3. Define clear tool descriptions so the agent can decide when and how to use them.
4. Implement an agent that:

- accepts a high-level project goal,
- creates and updates tickets using the MCP server,
- queries existing tickets to track progress.

5. Log and analyze the agent's tool usage and decision-making behavior.

Deliverables: A notebook demonstrating the agent, example project goals, generated tickets, and a brief analysis of agent behavior.

Exercise 3 AI assistant for IT services at Hagenberg campus

The university provides IT services required by students and employees alike. Information about these services is collected in an e-learning course and provided as a set of PDF documents. Finding relevant information in these documents can be quite cumbersome. You are therefore tasked with creating an LLM-powered assistant that answers questions about university IT services.

Because this information changes frequently, fine-tuning a language model is not a sustainable option. Instead, you will build a Retrieval-Augmented Generation (RAG) system that enables an LLM to answer questions based on the provided PDF documents.

An entrypoint to RAG with LangChain can be found [here](#).

Tasks:

1. Load and process the provided PDF files containing IT service information using LangChain.
2. Split the documents into suitable text chunks and generate embeddings for them.
3. Store the embeddings in a vector database that supports similarity search.
4. Implement a retrieval pipeline that selects relevant document chunks for a given user query.
5. Integrate a large language model via LangChain to generate answers grounded in the retrieved documents.
6. Demonstrate that the assistant answers questions based on the document content rather than relying on prior model knowledge.

Deliverables: A notebook that demonstrates the RAG pipeline, including PDF ingestion, vector store creation, retrieval, and question answering. The notebook should contain example queries with generated answers and a short discussion of the benefits and limitations of using RAG for this use case.

When using Large Language Models (e.g., ChatGPT, Copilot, ...) to produce output that is submitted, also state the prompt and model used in the submission. Submissions that are not marked as such and are noticeably AI-generated will be graded with zero points!

Submission: electronically via Moodle;
Deadline: Mon, May 28, 2026, 23:59.